TR-4384: SAP HANA on NetApp FAS Systems with Fibre Channel Protocol Configuration Guide

NetApp Solutions

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Nils Bauer and Marco Schoen, NetApp

The NetApp FAS product family has been certified for use with SAP HANA in TDI projects. The certified enterprise storage platform is characterized by the NetApp ONTAP operating system.

The certification is valid for the following models:

- FAS2720, FAS2750, FAS8200, FAS8300, FAS8700, FAS9000

For a complete list of NetApp's certified storage solutions for SAP HANA, see the certified and supported SAP HANA hardware directory.

This document describes FAS configurations that use the Fibre Channel Protocol (FCP).

> The configuration described in this paper is necessary to achieve the required SAP HANA KPIs and the best performance for SAP HANA. Changing any settings or using features not listed herein might result in performance degradation or unexpected behavior and should only be done if advised by NetApp support.

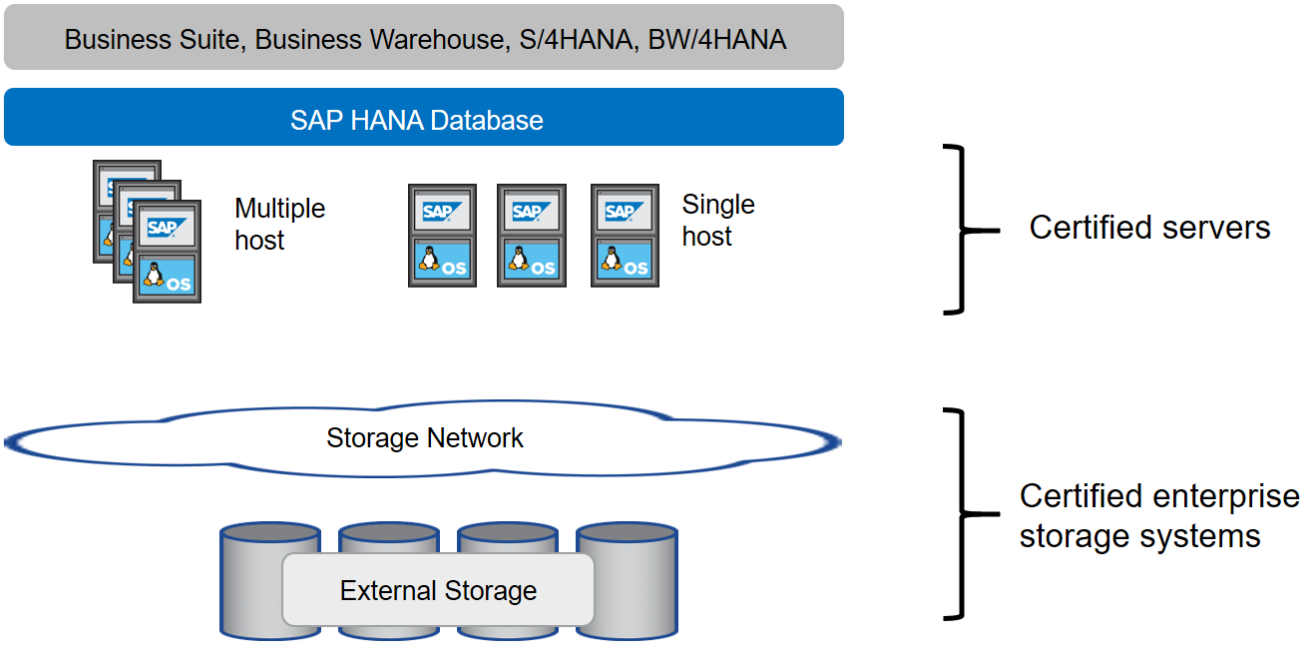
The configuration guides for FAS systems using NFS and NetApp AFF systems can be found using the following links:

- SAP HANA on NetApp AFF Systems with Fibre Channel Protocol
- SAP HANA on NetApp FAS Systems with NFS
- SAP HANA on NetApp AFF Systems with NFS

In an SAP HANA multiple-host environment, the standard SAP HANA storage connector is used to provide fencing in the event of an SAP HANA host failover. Refer to the relevant SAP notes for operating system configuration guidelines and HANA-specific Linux kernel dependencies. For more information, see SAP Note 2235581 – SAP HANA Supported Operating Systems.

SAP HANA tailored data center integration

NetApp FAS storage controllers are certified in the SAP HANA Tailored Data Center Integration (TDI) program using NFS (NAS) and Fibre Channel (SAN) protocols. They can be deployed in any SAP HANA scenario, such as, SAP Business Suite on HANA, S/4HANA, BW/4HANA or SAP Business Warehouse on HANA in single-host or multiple-host configurations. Any server that is certified for use with SAP HANA can be combined with the certified storage solution. See the following figure for an architecture overview.



For more information regarding the prerequisites and recommendations for productive SAP HANA systems, see the following resources:

- SAP HANA Tailored Data Center Integration Frequently Asked Questions
- SAP HANA Storage Requirements

SAP HANA using VMware vSphere

There are several options for connecting storage to virtual machines (VMs). The preferred one is to connect the storage volumes with NFS directly out of the guest operating system. This option is described in SAP HANA on NetApp AFF Systems with NFS.

Raw device mappings (RDM), FCP datastores, or VVOL datastores with FCP are supported as well. For both datastore options, only one SAP HANA data or log volume must be stored within the datastore for productive use cases. In addition, Snapshot- based backup and recovery orchestrated by SnapCenter and solutions based on this, such as SAP System cloning, cannot be implemented.

For more information about using vSphere with SAP HANA, see the following links:

- SAP HANA on VMware vSphere - Virtualization - Community Wiki
- Best Practices and Recommendations for Scale-Up Deployments of SAP HANA on VMware vSphere
- Best Practices and Recommendations for Scale-Out Deployments of SAP HANA on VMware vSphere
- 2161991 - VMware vSphere configuration guidelines - SAP ONE Support Launchpad (Login required)

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